

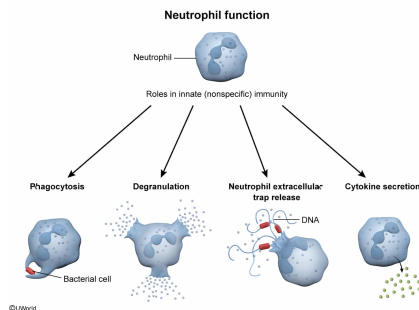
Assume that a patient who was originally infected with a certain strain of bacteria becomes reinfected with the same strain. Of the following cell types, which is LEAST likely to produce memory cells during the initial infection that would rapidly respond to the new infection?

- ☐ A. B cell [3%]
☐ B. Cytotoxic T cell [17%]
☐ C. Helper T cell [3%]
✓ ☒ D. Neutrophil [75%]

Correct

75% answered correctly

Explanation:



The immune system has **two primary subdivisions**:

- The **innate (nonspecific) immune system** comprises physical and chemical barriers (eg, **skin**, **mucous membranes**, **stomach acid**) that act to prevent pathogens from entering the body, as well as innate immune cells (eg, **neutrophils**, **macrophages**, **natural killer cells**) that can destroy pathogens and regulate the responses of other immune cells (ie, via the release of signaling molecules). Innate immune responses are directed at many types of pathogens (ie, are nonspecific) and can occur immediately upon initial contact with a pathogen.
- The **adaptive (specific) immune system** comprises cells that mount responses against specific pathogens via recognition of specific antigens (ie, foreign molecules). Adaptive immune cells include both **B cells** and **T cells** (eg, **cytotoxic T cells**, **helper T cells**). Upon initial exposure to a specific pathogen, some of these B and T cells divide and differentiate into specialized cells called **memory cells**. Memory B and T cells are stored in various tissues (eg, **lymphoid tissues**) and can **more rapidly** recognize and respond to future infections by the same pathogen.

This question states that a patient previously infected by a certain strain of bacteria has become reinfected with the same strain. *Memory B and T cells* produced in the response to the previous infection will be activated in this patient to combat the new infection (**Choices A, B, and C**).

Of the various **immune cell types**, neutrophils are the most numerous. They are innate (ie, nonspecific) immune cells that perform various functions, including the killing of pathogens via **phagocytosis** and the release of granules (ie, particles) that contain compounds (ie, antimicrobials) toxic to invading pathogens. Neutrophils also destroy pathogens by releasing DNA-containing material called neutrophil extracellular traps, and they release cytokines (chemical signals) that help regulate immune responses. However, unlike B and T cells of the adaptive immune system, innate neutrophils do *not* produce memory cells.

Educational objective:

Neutrophils are important components of the innate (nonspecific) immune system. Neutrophils perform many functions, including the destruction of pathogens via phagocytosis, the release of granules containing antimicrobial compounds, and the release of neutrophil extracellular traps containing DNA. Neutrophils also release cytokines (chemical signals) that regulate different immune responses.

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